Scenario 1

In this scenario, a new employee with the username `researcher9` joins an organization. You have to add them to the system and continue to manage their access during their time with the organization.
Here's how you'll do this task: **First**, you'll add a new employee to the system and then to their primary group. **Second**, you'll make this employee the owner of a file related to a particular project. **Third**, you'll add the new employee to a supplementary group. **Finally**, you'll delete the employee from the system.

```
analyst@503e1301b4d8:~$ sudo useradd researcher9
analyst@503e1301b4d8:~$ sudo usermod -g research_team researcher9
analyst@503e1301b4d8:~$ ls
analyst@503e1301b4d8:~$ sudo chown researcher9 /home/researcher2/projects
/project_r.txt
analyst@503e1301b4d8:~$ sudo usermod -a -G sales_team researcher9
analyst@503e1301b4d8:~$ sudo userdel researcher9
userdel: group researcher9 not removed because it is not the primary grou
p of user researcher9.
analyst@503e1301b4d8:~$ sudo groupdel researcher9
analyst@503e1301b4d8:~$
```

Scenario 2

In this scenario, you must examine and manage the permissions on the files in the `/home/researcher2/projects` directory for the `researcher2` user. The `researcher2` user is part of the `research_team` group.

You must check the permissions for all files in the directory, including any hidden files, to make sure that permissions align with the authorization that should be given. When it doesn't, you must change the permissions.
Here's how you'll do this task: **First**, you'll check the user and group permissions for all files in the `projects` directory. **Next**, you'll check whether any files have incorrect permissions and change the permissions as needed. **Finally**, you'll check the permissions of the `/home/researcher2/projects/drafts` directory and modify these permissions to remove any unauthorized access.

```
projects
researcher2@39f558891379:~$ cd projects
researcher2@39f558891379:~/projects$ ls
drafts  project_k.txt  project_m.txt  project_r.txt  project_t.txt
researcher2@39f558891379:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Mar 24 18:56 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Mar 24 18:56 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar 24 18:56 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_t.txt
researcher2@39f558891379:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 24 18:56 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 24 19:08 ..
-rw--w---- 1 researcher2 research_team   46 Mar 24 18:56 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 24 18:56 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Mar 24 18:56 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar 24 18:56 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_t.txt
researcher2@39f558891379:~/projects$ chmod o-w project_k.txt
researcher2@39f558891379:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Mar 24 18:56 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar 24 18:56 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_t.txt
researcher2@39f558891379:~/projects$ chmod g-r project_m.txt
researcher2@39f558891379:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Mar 24 18:56 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_k.txt
-rw------- 1 researcher2 research_team   46 Mar 24 18:56 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_t.txt
researcher2@39f558891379:~/projects$
```

```
drwxr-xr-x 3 researcher2 research_team 4096 Mar 24 19:08 ..
-rw--w---- 1 researcher2 research_team   46 Mar 24 18:56 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 24 18:56 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_k.txt
-rw------- 1 researcher2 research_team   46 Mar 24 18:56 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_t.txt
researcher2@39f558891379:~/projects$ chmod u-w,g-w,g+r .project_x.txt\
>
researcher2@39f558891379:~/projects$ ls -ls
total 20
4 drwx--x--- 2 researcher2 research_team 4096 Mar 24 18:56 drafts
4 -rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_k.txt
4 -rw------- 1 researcher2 research_team   46 Mar 24 18:56 project_m.txt
4 -rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_r.txt
4 -rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_t.txt
researcher2@39f558891379:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 24 18:56 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 24 19:08 ..
-r--r----- 1 researcher2 research_team   46 Mar 24 18:56 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 24 18:56 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_k.txt
-rw------- 1 researcher2 research_team   46 Mar 24 18:56 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_t.txt
researcher2@39f558891379:~/projects$ chmod g-x drafts
researcher2@39f558891379:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 24 18:56 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 24 19:08 ..
-r--r----- 1 researcher2 research_team   46 Mar 24 18:56 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Mar 24 18:56 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_k.txt
-rw------- 1 researcher2 research_team   46 Mar 24 18:56 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 24 18:56 project_t.txt
researcher2@39f558891379:~/projects$
```

Scenario 3

In this scenario, you need to ensure that the `/home/analyst` directory is properly organized. You have to make a few changes to the `/home/analyst` directory and the files it contains. You also have to edit a file to record the changes or updates you make to the directory.

- Create a new directory
- Remove a directory
- Move a file and delete a file
- Create a file and add text using nano

```
analyst@d46ac883f828:~$ pwd
/home/analyst
analyst@d46ac883f828:~$ ls
notes  reports  temp
analyst@d46ac883f828:~$ mkdir logs
analyst@d46ac883f828:~$ ls
logs  notes  reports  temp
analyst@d46ac883f828:~$ rmdir temp
analyst@d46ac883f828:~$ ls
logs  notes  reports
analyst@d46ac883f828:~$ cd notes
analyst@d46ac883f828:~/notes$ ls
Q3patches.txt  tempnotes.txt
analyst@d46ac883f828:~/notes$ mv Q3patches.txt /home/analyst/reports
analyst@d46ac883f828:~/notes$ ls
tempnotes.txt
analyst@d46ac883f828:~/notes$ cd -
/home/analyst
analyst@d46ac883f828:~$ cd reports
analyst@d46ac883f828:~/reports$ ls
Q1patches.txt  Q2patches.txt  Q3patches.txt
analyst@d46ac883f828:~/reports$ cd notes
-bash: cd: notes: No such file or directory
analyst@d46ac883f828:~/reports$ cd -
/home/analyst
analyst@d46ac883f828:~$ cd notes
analyst@d46ac883f828:~/notes$ ls
tempnotes.txt
analyst@d46ac883f828:~/notes$ cat tempnotes.txt
This file no longer contains any useful data and can be removed.
analyst@d46ac883f828:~/notes$ rm tempnotes.txt
analyst@d46ac883f828:~/notes$ ls
analyst@d46ac883f828:~/notes$ touch new_tempnote.txt
analyst@d46ac883f828:~/notes$ ls
new_tempnote.txt
analyst@d46ac883f828:~/notes$ nano new_tempnotes.
```

```
GNU nano 5.4                    new_tempnotes.txt
This file longer contains any useful data and should not be removed at
the moment.
```

```
analyst@d46ac883f828:~/notes$ ls
new_tempnote.txt  new_tempnotes.txt
analyst@d46ac883f828:~/notes$ cat new_tempnote.txt
analyst@d46ac883f828:~/notes$ cat new_tempnotes.txt
This file longer contains any useful data and should not be removed at
the moment.
analyst@d46ac883f828:~/notes$ rm new_tempnote.txt
analyst@d46ac883f828:~/notes$ ls
new_tempnotes.txt
analyst@d46ac883f828:~/notes$
```